25900894 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR/IMPROVEMENT/REAPPEARANCE EXAMINATION NOVEMBER 2025

FIRST SEMESTER

Discipline Specific Core Course (DSC) - MG1DSCMAT100 - GROUND ROOTS OF MATHEMATICS

(2024 ADMISION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Type Questions

Answer any **5** Questions

Each Question carries **2** marks

1. Find the predicate in the statement: x spends more than five hours every weekday in class. [U] / [CO1]
2. Let P(x) denote the statement $x > 3$. What is the truth value of the quantification $\exists x P(x)$, where the domain consists of all real numbers? [U] / [CO1]
3. Find the inverse of $f(x) = \sqrt{x+5}$. [A] / [CO3]
4. What is the greatest integer function? [K] / [CO2]
5. Find $\frac{dy}{dx}$ if $y = x \sin x$ [A] / [CO4]
6. Find $\frac{dy}{dx}$ if $x\ cosy = y$. [A] / [CO4]
7. Find the inflection points of $f(x) = x^3 - 3x^2 + 1$. [U] / [CO5]
8. Find $\lim_{x\to\infty} \frac{\ln x}{2\sqrt{x}}$ [A] / [CO6]

[5x2 = 10]

Part B

Short Essay Type Questions

Answer any **5** Questions

Each Question carries **6** marks

9. Let p and q be the propositions. p : It is below freezing q: It is snowing. [U] / [CO1]
   Write these propositions using p and q and logical connectives.
   a) It is below freezing and snowing.
   b) It is not below freezing and it is not snowing.
   c) If it is below freezing, it is also snowing.

10. Prove or disprove: $\forall x P(x) \vee \forall x Q(x)$ and $\forall x(P(x) \vee Q(x))$ are logically equivalent. [U] / [CO1]

11. Given the function $y = x^2$, write the formula when the graph is shifted [U] / [CO2]
    (i) 1 unit up
    (ii) 2 units down.

12. What is a one-to-one function and what test can be used to determine if a function is one-to-one? Determine from its graph if the function $y = cosx$ is one-to-one or not. [A] / [CO3]

13. Find $F'(1)$ if $F(x) = \frac{f(x)}{g(x)}$ given that $f(1) = -1, f'(1) = 2, g(1) = 3$ and $g'(1) = -1$. [A] / [CO4]

14. Find $\frac{dx}{dt}$, if $x = \frac{t^2+1}{3t}$. [A] / [CO4]

15. Find all critical points of $f(x) = 3x^{5/3} - 15x^{2/3}$. [U] / [CO5]

16. Identify the type of indeterminate form and find $\lim_{x\to 0} (1+2x)^{-3/x}$ [A] / [CO6]

[5x6 =30]

Part C

Essay Type Questions

Answer any **3** Questions

Each Question carries **10** marks

17. a) Show that $(p \wedge q) \rightarrow (p \vee q)$ is a tautology. [U] / [CO1]
    b) Show that $\neg(p \rightarrow q)$ and $p \wedge \neg q$ are logically equivalent.

18. a) Define even and odd functions. [U] / [CO2]
    b) Show that the function $f(x) = 0$ is an odd function as well as an even function.

19. (a) Let $f(x) = \sqrt{x}$ and $g(x) = \sqrt{1-x}$. Find $\frac{f}{g}$ and $\frac{g}{f}$. Also, write the domain of each of the functions. [U] / [CO2]
    (b) Given the function $y = x^3$, write the formula when the graph of the function is shifted 1 unit to the left and 1 unit down.

20. (a) Find the derivative of $y = (3x^2 + 2x - 1)^6$. [A] / [CO4]

(b) Find $\frac{dy}{dx}$ if $y = (x^2 \tan x)^4$.

21. Suppose that $f(x)$ has derivative $f'(x) = x(x-4)^2$ and [U] / [CO5]
$f''(x) = (x-4)(3x-4)$.

(a) Find the intervals in which the function is increasing.

(b) Find the intervals in which the function is concave up.

(c) Find the intervals in which the function is concave down.

22. Evaluate the following limits using L'Hôpital's rule. [A] / [CO6]

(a) $\lim_{t \to 0} \frac{1-e^t}{te^t}$

(b) $\lim_{x \to +\infty} \frac{x^{-\frac{4}{3}}}{\sin(\frac{1}{x})}$.

[3x10 =30]